

Leray Boundedness Without Uniform Continuity

A full candidate solution to the omitted case in arXiv:2405.12753

Automated mathematics run `fa_banach.001`, lane 11

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Status. Full solution candidate, likely valid, awaiting human review. For the source class of convex Reinhardt domains, the proof removes the uniform-continuity hypothesis and completes the Leray-transform dichotomy: boundedness holds exactly when $\kappa_3/(\kappa_1\kappa_2)$ is bounded above and bounded away from zero.

1 Source problem

The source is A. Chatterjee, *The Laplace and Leray transforms on some (weakly) convex domains in $\mathbb{C}^2$* , arXiv:2405.12753 [1]. Its Theorem 1.4 considers the densely defined Leray transform $\mathbb{L}_b$ on a domain $\Omega \in \widetilde{\mathcal{R}}$, with respect to the canonical boundary Monge–Ampère measure μ_Ω . It proves:

1. boundedness if the curvature ratio

$$K_\Omega = \frac{\kappa_3}{\kappa_1\kappa_2}$$

is uniformly continuous and bounded away from zero; and

2. unboundedness if K_Ω is not bounded above or is not bounded away from zero.

The sole remaining case is therefore that K_Ω is bounded above and away from zero but has no continuous extension to the axes. Section 4.3 of the current source PDF isolates this case and explains why the regular variation argument used in the positive theorem no longer applies.

In the source parametrization,

$$\begin{aligned} r_1(s) &= b_1 \exp\left(-\int_s^1 \frac{dt}{t\check{p}(t)}\right), \\ r_2(s) &= b_2 \exp\left(-\int_0^s \frac{dt}{(1-t)\check{p}(t)}\right), \quad 0 < s < 1, \end{aligned}$$

where $\check{p} : (0, 1) \rightarrow (1, \infty)$ is the exponent function. The dual domain has

$$r_1^*(s) = \frac{s}{r_1(s)}, \quad r_2^*(s) = \frac{1-s}{r_2(s)}.$$

The source notes that its omitted geometric case makes $\check{p}$ remain bounded away from 1 and infinity but permits it to oscillate without an endpoint limit. We prove boundedness using only those two bounds.

$$C_{k,l} = \inf_{j \in \mathbb{N}} \left(\frac{2k}{\check{p}(s_{m_j, n_j})} + 1 - k \right) > 0.$$

These are the necessary modifications of [5, (5.4)] and [5, (5.5)] needed in our case. As in the case of [5, (5.13), (5.17), (5.19)], it follows that, for all $k \in \{-1, 0, 1\}$,

- (1) $\lim_{j \rightarrow \infty} A_{m_j, n_j, k} / s_{m_j, n_j} = \lim_{j \rightarrow \infty} A_{m_j, n_j, k} / (1 - s_{m_j, n_j}) = 0$,
- (2) $\lim_{j \rightarrow \infty} g_{m_j, n_j, k}(t) = e^{-t^2}$ for all $t \in \mathbb{R}$, and
- (3) $g_{m_j, n_j, k}(t) \leq e^{C_{k,l}(1-2^{-|k|/2}|t|)}$ for all $t \in \mathbb{R}$, when j is large enough.

By the dominated convergence theorem, (4.15) holds with $c = \sqrt{\pi}$. Combining this with (4.14), we obtain (4.13).

4.3. Remark. The case where $\kappa_3(\kappa_1\kappa_2)^{-1}$ is bounded away from zero on $b\Omega_+$ but not uniformly continuous on $b\Omega_+$ is quite subtle. This condition implies that the exponent $\check{p} : (0, 1) \rightarrow (1, \infty)$ remains bounded away from 1 and ∞ , but fails to be uniformly continuous on $(0, 1)$. Without loss of generality, we assume that the limit of $\check{p}(s)$ does not exist as $s \rightarrow 0$. Under this assumption, the key difficulty in proving the $L^2(\mu_\Omega)$ -boundedness of $\mathbb{L}_b$ lies in verifying conditions (4.4) and (4.5). The crux of the issue is that r_1 and r_1^* lose the property of being regularly varying at 0 with a positive index. This follows from Karamata's representation theorem (see [37, Theorem 1.2]), which says that if $f : (0, 1) \rightarrow \mathbb{R}$ is a regularly varying function at 0, with index $\ell > 0$, then

$$f(s) = s^\ell \exp \left(c(s) + \int_s^1 \frac{\epsilon(t)}{t} dt \right), \quad s \in (0, 1),$$

where c, ϵ are some bounded measurable functions with, $\lim_{s \rightarrow 0} c(s) = c$, $\lim_{s \rightarrow 0} \epsilon(s) = 0$. However, from (3.3) and (3.11), it follows that such a representation would hold for r_1 or r_1^* only if $\check{p}(s)$ has a finite limit as $s \rightarrow 0$, which contradicts our assumption. On the other hand, if we attempt to prove that $\mathbb{L}_b$ does not extend as an $L^2(\mu_\Omega)$ -bounded operator by adapting the technique in Theorem 1.4.ii, the attempt fails as (4.13) continues to hold. Since $\check{p}$ is bounded away from 1, and ∞ on $(0, 1)$, $\sqrt{\check{p}\left(\frac{x_l}{1+x_l}\right)\check{p}^*\left(\frac{x_l}{1+x_l}\right)}$ remains bounded above for all $x_l \in (0, \infty)$.

5. PROOF OF THEOREM 1.5

Figure 1: The exact omitted case, source Section 4.3, page 20.

2 Proof intuition

The source analyzes each monomial mode by endpoint regular variation. That is more asymptotic information than boundedness needs. For mode (m, n) , the primal density

$$F = r_1^{2m} r_2^{2n}$$

and its dual density

$$F^* = (r_1^*)^{2m} (r_2^*)^{2n}$$

have their maximum at the same point $s_0 = m/(m+n)$. Moreover, $FF^* = s^{2m}(1-s)^{2n}$, the ordinary beta kernel.

The logarithmic derivative of F is $1/\check{p}(s)$ times the logarithmic derivative of that beta kernel; for F^* , the multiplier is $1 - 1/\check{p}(s)$. Uniform upper and lower bounds on $\check{p}$ therefore dominate each normalized density by a fixed positive power of the beta kernel. Its integral has width $\sqrt{(m+1)(n+1)/(m+n+1)^{3/2}}$. The product of the primal and dual widths cancels exactly against the factorial coefficient in the source rank-one norm formula. This gives a uniform mode bound without requiring any limit of $\check{p}$.

3 Source rank-one criterion

The source decomposes $L^2(b\Omega, \mu_\Omega)$ orthogonally into monomial subspaces. Its Lemma 3.5 states that, for $m, n \geq 0$,

$$\|\mathbb{L}_{m,n}\|_{\mu_\Omega}^2 = \gamma_{m,n}^2 \left(\int_0^1 r_1(s)^{2m} r_2(s)^{2n} ds \right) \left(\int_0^1 r_1^*(s)^{2m} r_2^*(s)^{2n} ds \right), \quad (1)$$

where

$$\gamma_{m,n} = \frac{(m+n+1)!}{m!n!}.$$

It also states that $\mathbb{L}_b$ is bounded if and only if these rank-one norms are uniformly bounded, in which case $\mathbb{L}_b$ is a bounded projection onto $\mathcal{H}^2(\Omega, \mu_\Omega)$.

where the above series converges uniformly in z on compact subsets of Ω .

Following [5], a function of the form $g(s)e^{i(m_1\theta_1+m_2\theta_2)}$ is called an (m_1, m_2) -monomial, and the space of all (m_1, m_2) -monomials that are in $L^2(b\Omega, \mu_\Omega)$ is denoted by $L_{m_1, m_2}^2(b\Omega, \mu_\Omega)$. The spaces $L_{m_1, m_2}^2(b\Omega, \mu_\Omega)$, $(m_1, m_2) \in \mathbb{Z} \times \mathbb{Z}$, are mutually orthogonal, and

$$L^2(b\Omega, \mu_\Omega) = \bigoplus_{(m_1, m_2) \in \mathbb{Z}^2} L_{m_1, m_2}^2(b\Omega, \mu_\Omega). \quad (3.25)$$

By direct computation, it is shown in [5, Corollary 23] that $\mathbb{L}$ and $\mathbb{L}_b$ are well-defined on each $L_{m_1, m_2}^2(b\Omega, \mu_\Omega)$, and therefore densely-defined on $L^2(b\Omega, \mu_\Omega)$. Furthermore, $\mathbb{L}_b$ reproduces (the restrictions to $b\Omega$ of) holomorphic polynomials, and annihilates any (m_1, m_2) -monomial with $\min\{m_1, m_2\} < 0$. Finally, since the measure μ_Ω is admissible in the sense of [5, Definition 28], we have the following result as a combination of Proposition 26, Theorem 30 and Proposition 32 in [5].

Lemma 3.5. *Let $\Omega \in \tilde{\mathcal{R}}$. Given $(m_1, m_2) \in \mathbb{Z} \times \mathbb{Z}$, let $\mathbb{L}_{m_1, m_2}$ denote the restriction of $\mathbb{L}_b$ to $L_{m_1, m_2}^2(b\Omega, \mu_\Omega)$.*

- (1) *If $\min\{m_1, m_2\} \geq 0$, then $\mathbb{L}_{m_1, m_2}$ is a bounded rank-one projection operator on $L_{m_1, m_2}^2(b\Omega, \mu_\Omega)$, with*

$$\|\mathbb{L}_{m_1, m_2}\|_{\mu_\Omega}^2 = \gamma_{m_1, m_2}^2 \int_0^1 r_1(s)^{2m_1} r_2(s)^{2m_2} ds \int_0^1 r_1^*(s)^{2m_1} r_2^*(s)^{2m_2} ds, \quad (3.26)$$

where $\gamma_{m_1, m_2} = \frac{(m_1+m_2+1)!}{m_1!m_2!}$.

- (2) *The densely-defined operator $\mathbb{L}_b$ extends as a bounded operator on $L^2(b\Omega, \mu_\Omega)$ if and only if the quantities $\|\mathbb{L}_{m_1, m_2}\|_{\mu_\Omega}^2$ are uniformly bounded for $(m_1, m_2) \in \mathbb{N} \times \mathbb{N}$.*
- (3) *If $\mathbb{L}_b$ extends as a bounded operator on $L^2(b\Omega, \mu_\Omega)$, it extends as a bounded projection operator from $L^2(b\Omega, \mu_\Omega)$ onto $\mathcal{H}^2(\Omega, \mu_\Omega)$.*

4. PROOF OF THEOREM 1.4

For the first claim, we essentially mimic the proof of Theorem 1 in [5]. The main technical tool used therein is Theorem 45, which gives the asymptotic values of the operator-norms of $\mathbb{L}_{m_1, m_2}$

Figure 2: The rank-one norm formula and boundedness criterion, source Lemma 3.5, page 16.

4 Two uniform estimates

We use the conventions $0^0 = 1$ and $0 \log 0 = 0$.

Lemma 1 (Normalized beta width). *For each fixed $c > 0$, there is $C_c < \infty$ such that, for all $m, n \geq 0$, $M = m + n > 0$, and*

$$B_{m,n}(s) = s^{2m}(1-s)^{2n}, \quad s_0 = \frac{m}{M},$$

one has

$$J_c(m, n) := \int_0^1 \left(\frac{B_{m,n}(s)}{B_{m,n}(s_0)} \right)^c ds \leq C_c \frac{\sqrt{(m+1)(n+1)}}{(M+1)^{3/2}}. \quad (2)$$

Proof. The beta integral gives the exact identity

$$J_c(m, n) = \frac{B(2cm+1, 2cn+1)}{[(m/M)^{2m}(n/M)^{2n}]^c}. \quad (3)$$

Suppose first that $m, n \geq 1$. The standard two-sided Stirling inequalities, applied to the three gamma factors in

$$B(2cm+1, 2cn+1) = \frac{\Gamma(2cm+1)\Gamma(2cn+1)}{\Gamma(2cM+2)},$$

give, with constants depending only on c ,

$$B(2cm+1, 2cn+1) \leq C_c \frac{m^{2cm+1/2} n^{2cn+1/2}}{M^{2cM+3/2}}.$$

Dividing by the denominator in (3) yields

$$J_c(m, n) \leq C_c \frac{\sqrt{mn}}{M^{3/2}},$$

which implies (2).

If $m = 0$ and $n = M > 0$, then

$$J_c(0, M) = \int_0^1 (1-s)^{2cM} ds = \frac{1}{2cM+1} \leq \frac{C_c}{M+1}.$$

This is exactly (2) on that axis. The case $n = 0$ is symmetric. $\square$

Lemma 2 (Factorial-height estimate). *There is an absolute constant C such that, for all $m, n \geq 0$ and $M = m + n > 0$,*

$$\gamma_{m,n}^2 B_{m,n}(s_0) \leq C \frac{(M+1)^3}{(m+1)(n+1)}. \quad (4)$$

Proof. Since $\gamma_{m,n} = (M+1) \binom{M}{m}$, the two-sided Stirling inequalities give, when $m, n \geq 1$,

$$\binom{M}{m} \left(\frac{m}{M} \right)^m \left(\frac{n}{M} \right)^n \leq C \sqrt{\frac{M}{mn}}.$$

Squaring and multiplying by $(M+1)^2$ proves (4) in the interior, after harmlessly replacing m, n, M by $m+1, n+1, M+1$. If $m = 0$, then $\gamma_{0,M} = M+1$ and $B_{0,M}(0) = 1$, so both sides of (4) are comparable to $(M+1)^2$. The other axis is symmetric. $\square$

5 Main theorem

Theorem 3 (Bounded exponent implies bounded Leray projection). *Let $\Omega \in \widetilde{\mathcal{R}}$. Suppose its exponent function satisfies*

$$1 < p_- \leq \check{p}(s) \leq p_+ < \infty \quad (0 < s < 1). \quad (5)$$

No endpoint limits or uniform continuity are assumed. Then $\mathbb{L}_b$ extends to a bounded projection

$$L^2(b\Omega, \mu_\Omega) \longrightarrow \mathcal{H}^2(\Omega, \mu_\Omega).$$

Quantitatively, its norm is bounded by a constant depending only on p_- and p_+ .

Proof. Fix $m, n \geq 0$, set $M = m + n$, and suppose first that $M > 0$. Write

$$\begin{aligned} F(s) &= r_1(s)^{2m} r_2(s)^{2n}, \\ F^*(s) &= r_1^*(s)^{2m} r_2^*(s)^{2n}, \\ B(s) &= B_{m,n}(s) = s^{2m} (1-s)^{2n}. \end{aligned}$$

The dual identities give

$$F(s)F^*(s) = B(s). \quad (6)$$

Differentiating the source integral formulas for r_1, r_2 gives

$$(\log F)'(s) = \frac{2}{\check{p}(s)} \left(\frac{m}{s} - \frac{n}{1-s} \right), \quad (7)$$

$$(\log F^*)'(s) = 2 \left(1 - \frac{1}{\check{p}(s)} \right) \left(\frac{m}{s} - \frac{n}{1-s} \right), \quad (8)$$

$$(\log B)'(s) = 2 \left(\frac{m}{s} - \frac{n}{1-s} \right). \quad (9)$$

Thus all three functions increase before $s_0 = m/M$ and decrease after s_0 , with the evident one-sided interpretation on the axes.

Put

$$a = \frac{1}{p_+} > 0, \quad b = 1 - \frac{1}{p_-} > 0.$$

On either side of s_0 , integrate (7)–(9), taking account of the common sign of $m/s - n/(1-s)$. Since $1/\check{p} \geq a$, one obtains

$$\frac{F(s)}{F(s_0)} \leq \left(\frac{B(s)}{B(s_0)} \right)^a. \quad (10)$$

Likewise $1 - 1/\check{p} \geq b$ gives

$$\frac{F^*(s)}{F^*(s_0)} \leq \left(\frac{B(s)}{B(s_0)} \right)^b. \quad (11)$$

Consequently,

$$\int_0^1 F \, ds \leq F(s_0) J_a(m, n), \quad \int_0^1 F^* \, ds \leq F^*(s_0) J_b(m, n).$$

Using (6), Lemmas 1 and 2, and the source norm formula (1), we obtain

$$\begin{aligned}\|\mathbb{L}_{m,n}\|_{\mu_\Omega}^2 &\leq \gamma_{m,n}^2 B(s_0) J_a(m,n) J_b(m,n) \\ &\leq C \frac{(M+1)^3}{(m+1)(n+1)} \cdot C_a C_b \frac{(m+1)(n+1)}{(M+1)^3} \\ &\leq C_{p-,p+}.\end{aligned}$$

For $m = n = 0$, formula (1) gives norm one. Hence all rank-one norms are uniformly bounded. The source's Lemma 3.5 now implies the asserted bounded projection. $\square$

Corollary 4 (Complete geometric dichotomy). *Let $\Omega \in \widetilde{\mathcal{R}}$. Then $\mathbb{L}_b$ is bounded on $L^2(b\Omega, \mu_\Omega)$ if and only if*

$$\frac{\kappa_3}{\kappa_1 \kappa_2}$$

is bounded above and bounded away from zero on $b\Omega \setminus \mathcal{Z}$. Thus uniform continuity in part (i) of the source's Theorem 1.4 is unnecessary.

Proof. Let $K(s)$ denote the curvature ratio in the source parametrization. Its curvature identity is

$$\check{p}(s) - 1 = K(s)Q(s), \quad Q(s) = \sqrt{\left(\frac{s}{r_1(s)}\right)^2 + \left(\frac{1-s}{r_2(s)}\right)^2}. \quad (12)$$

Because $\check{p} > 1$, the integral formulas for r_1, r_2 imply

$$b_1 s < r_1(s) \leq b_1, \quad b_2(1-s) < r_2(s) \leq b_2.$$

It follows that

$$\frac{1}{\sqrt{b_1^2 + b_2^2}} \leq Q(s) \leq \sqrt{\frac{1}{b_1^2} + \frac{1}{b_2^2}}.$$

Therefore K is bounded above and away from zero exactly when $\check{p}$ is bounded above and away from one. The forward implication now follows from Theorem 3. The reverse implication is part (ii) of the source's Theorem 1.4. $\square$

6 Verification

Symbolic audit. The proof uses only the source parametrization, duality identities, and rank-one criterion. The logarithmic derivatives in (7)–(9) were recomputed directly. The direction of (10) and (11) was checked separately on $s < s_0$ and $s > s_0$, where the shared derivative changes sign. The modes on both coordinate axes were treated explicitly in both Stirling lemmas.

Computational sanity check. The script `code/verify_beta_kernel_bounds.py` evaluates the exact log-beta and log-gamma expressions for every $(m, n) \in \{0, \dots, 300\}^2$, and for $c \in \{0.05, 0.1, 0.25, 0.5, 0.9\}$. It checks that the normalized quantities appearing in Lemmas 1 and 2 stay finite and records their maxima. This finite computation is a check, not part of the proof.

Artifact audit. The source gap and rank-one criterion were rendered directly from the official arXiv PDF. The packet was compiled with a clean LaTeX build, and every rendered packet page was inspected for clipping and legibility.

7 Scope, limitations, and novelty

The result fully answers the precise Leray boundedness case isolated in Section 4.3 and completes Theorem 1.4’s dichotomy on the source class $\widetilde{\mathcal{R}}$, for the boundary Monge–Ampère measure μ_Ω . It does not answer the paper’s broader introductory question for arbitrary weakly convex domains in higher dimension, nor does it address unrelated boundary measures.

A bounded novelty search was performed on 23 July 2026 using the arXiv id, the exact title, the phrase “bounded away from zero but not uniformly continuous,” and combinations of “Leray transform,” “Reinhardt,” and “exponent $p(s)$.” Search results found the source (including its 2025 journal publication), Barrett–Lanzani’s earlier rank-one theory, and related Leray-transform papers, but no later paper claiming to settle this omitted case. Novelty is therefore plausible but not certified.

Human-review recommendation. This is a high-priority full candidate. The main points to verify are: (i) the sign-sensitive integration producing (10)–(11); (ii) the uniform beta estimate when one index is zero; and (iii) the final use of the source rank-one boundedness criterion.

References

- [1] A. Chatterjee, *The Laplace and Leray transforms on some (weakly) convex domains in $\mathbb{C}^2$* , arXiv:2405.12753; J. Fourier Anal. Appl. **31** (2025), Article 36.
- [2] D. E. Barrett and L. Lanzani, *The spectrum of the Leray transform for convex Reinhardt domains in $\mathbb{C}^2$* , J. Funct. Anal. **257** (2009), 2780–2819.